

Intelligent VAPT Report Analysis and Remediation System

Introduction

Government organizations regularly perform Vulnerability Assessment and Penetration Testing (VAPT) to identify security weaknesses. The generated reports are often technical and difficult for developers to interpret. This project proposes an on-premise AI-assisted system that analyzes VAPT reports, explains vulnerabilities, recommends remediation steps, and provides secure coding guidance without relying on external cloud services.

Objective of the Project

- Parse uploaded VAPT reports automatically.
- Extract vulnerability details.
- Map vulnerabilities with OWASP and CWE knowledge.
- Provide developer-friendly explanations.
- Recommend remediation and secure coding practices.
- Generate structured remediation reports.
- Operate entirely within the organization's infrastructure.

Scope of the Project

- Accept PDF VAPT reports.
- Extract vulnerability information.
- Match findings against a local knowledge base.
- Display explanations and remediation guidance.
- Generate downloadable reports.
- Assist developers after VAPT; does not perform scanning or penetration testing.

Technologies to be Used

Frontend: React.js, Tailwind CSS

Backend: FastAPI

Programming Language: Python

Database: PostgreSQL

PDF Parsing: PyMuPDF, pdfplumber

Knowledge Base: OWASP Top 10, CWE, CAPEC, Secure Coding Guidelines

Authentication: JWT

Report Generation: ReportLab

Deployment: Docker (On-Premise)

Methodology

1. User uploads a VAPT PDF.
2. Backend extracts vulnerability information.
3. System matches vulnerabilities against a locally stored knowledge base.
4. System returns predefined remediation steps, secure coding practices, and sample code.

5. Results are displayed through the web interface.
6. User downloads a structured remediation report.

Source of Data

- OWASP Top 10
- Common Weakness Enumeration (CWE)
- CAPEC Database
- CERT Secure Coding Standards
- NIST Secure Software Development Framework (SSDF)
- Sample VAPT reports from CDAC/authorized environments
- Organization-specific security guidelines

Implementation Site

To be deployed on CDAC Delhi or any government organization's on-premise infrastructure. The application will execute entirely within the internal network without transmitting sensitive data to external cloud services.

Expected Outcome

- Automated VAPT report analysis
- Easier vulnerability understanding
- Standardized remediation recommendations
- Secure coding guidance
- Developer-friendly reports
- Faster remediation process

Advantages

- On-premise deployment
- No dependence on cloud AI services
- Suitable for government organizations
- Faster vulnerability remediation
- Improved secure coding awareness
- Modular and scalable architecture

Conclusion

The proposed system simplifies the interpretation of VAPT reports by automatically extracting vulnerabilities and providing standardized explanations, remediation guidance, and secure coding recommendations. Running entirely within an organization's infrastructure ensures data confidentiality while supporting secure software development.